The Short, the Long, and the Consequences of VAR Analysis For Structural DSGE Estimation

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Motivation

- VAR models can be used to estimate DSGE frameworks by IRFs minimization matching
- If the VAR is mispecified the DSGE might also be so

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- *"If the VAR adequately caputeres the DGP, IRFs are optimal at all horizons.*
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  If the ***VAR is mispecified****, however, then the* ***specification errors will be computed at each horizon****"* **Ramey2016), p. 83**
- VARs and DSGEs are instruments used by central banks to make policy decisions
- Misspecification can deliver distorted results mis informing the central banker

This Paper

Research Question:

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What we do:

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Findings:

- Both VAR based IRFs and structural DSGE parameter estimates are significantly affected by the inclusion of longer lags

VAR IRFs DSGE IRFs DSGE Parameters

Related Literature

IRFs:

- (Plagborg-Møller and Wolf2021; Ludwig2026)

DSGE IRFs matching with VARs

- (Christiano et al.2005; Altig et al.2011; Jarociński and Karadi2020)

Truncation Bias

- (De Graeve and Westermark2025; Baumeister2025)

Long Lag VARs

- (Antolín-Díaz and Surico2025)

Illustration of Truncation Bias Problem

$$Y_t = B_1 Y_{t-1} + \cdots + B_p Y_{t-p} + u_t, \qquad E(u_t u_t') = \Sigma \tag{1}$$

$$B(L) = B_1 L + \cdots + B_p L^p \tag{2}$$

- The VAR with $p < p^*$ suffers from truncation bias.[1]
- Lag truncation biases the reduced form coefficients $B(L)$ and the reduced form variance covariance matrix Σ, which are the two main ingredients for IRF estimation.

$$\text{IRF} = f\big(B(L), \Sigma\big) \tag{3}$$

[1] Where p^* denotes the DGP true lag length.

Long-Lag

De Graeve and Westermark2025) argue that increasing the lag structure improves IRFs estimates obtaining:

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Lower Variance

- Misspecification reduction lowers the variance of the residuals
- This variance reduction counteracts the imprecision caused by increased parametrisation

Long-Lag

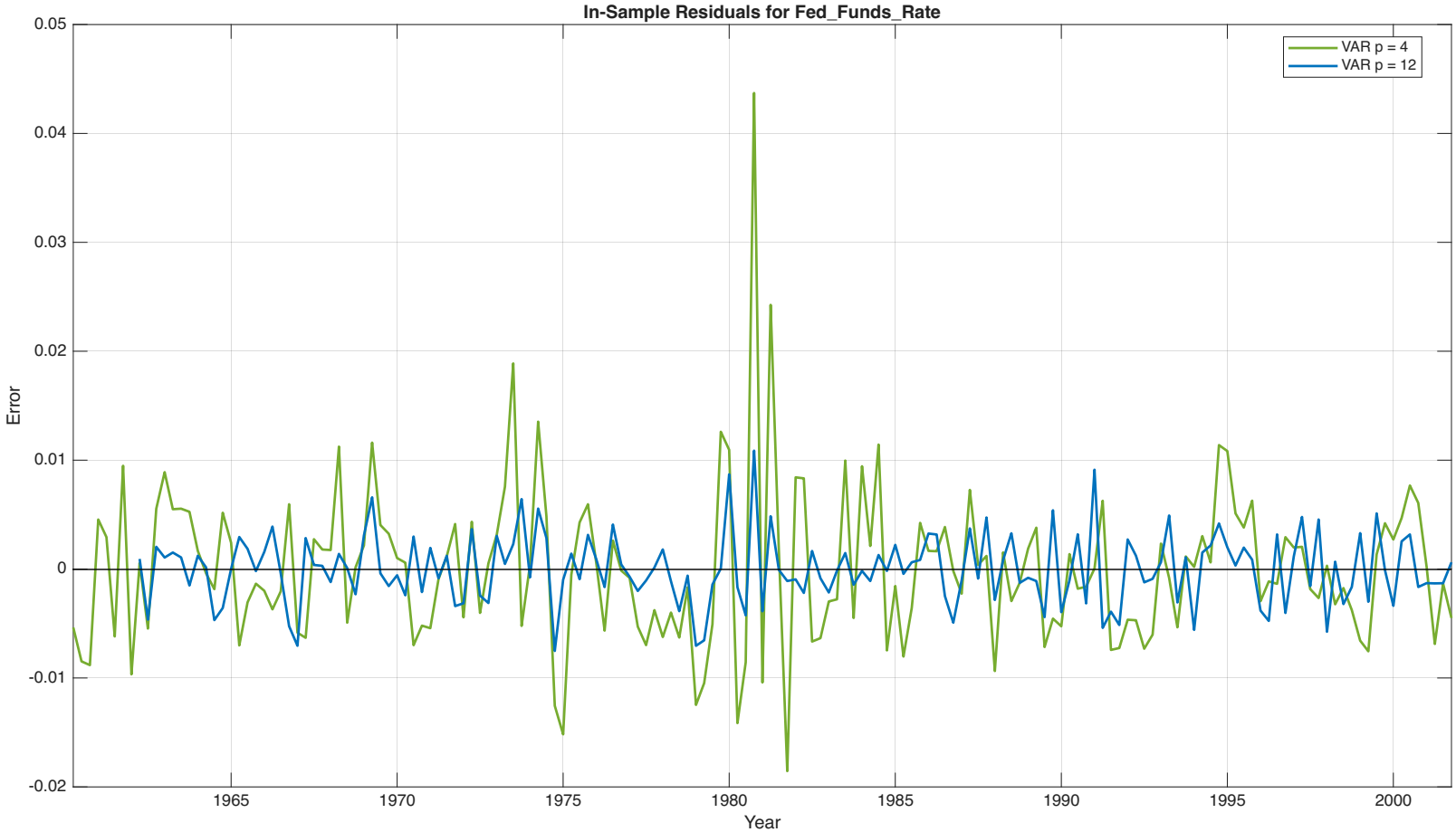


Figure 1: FFR Residuals Comparison Across lag 4 Vs 12

Long-Lag Benefits

The variance-reducing effect of lower misspecification often dominates the penalty of increased parameterisation.

Compared to short-lag VARs, Long-Lag VARs have:

- Reduced truncation bias
- More precise inference
- Lower MSE
- Better coverage

Baseline Exercise

Framework:

- **Altig et al.2011**)'s DSGE[2]
- Sample period 1959:Q2-2001:Q4

Exercise:

- We re-estimate the VAR with different lag order
- Re-estimate the DSGE by IRFs matching
- Analyse possible variations in the structural parameters across different lag choice
- Search for possible drivers of structural changes

[2]Where the replication package provided by the authors refers to the 2005 NBER working paper

ACEL2011's Mechanism

Classic NK models accountes for inflation inertia by assuming that firms re-optimise their prices every six quarter or even less often

- Altig et al.2011)'s model inflation is inertial despite firms re-optimise prices average every 1.8 quarters
- Inertia is explained by the fact that, when firms re optimise they do it by a small amount because each firm's short run marginal cost is increasing in its own output
- The key assumption is that, in any given period a firm's capital stock is pre-determined and is firm specific

They create a two model that builds on Christiano et al.2005):

- Capital is Homogeneous
- Capital Firm-Specific

ACEL2011's Mechanism (Cond't)

- When log-linearized, the only difference in characterizing equilibrium equations lies in the Phillips Curve equation

$$\Delta\hat{\pi}_t = \mathbb{E}[\beta\Delta\hat{\pi}_{t+1} + \gamma\hat{s}_t|\Omega_t] \tag{4}$$

$$\gamma = \frac{(1-\xi_p)(1-\beta\xi_p)}{\xi_p}\chi \tag{5}$$

- The s_t are the economy-wide marginal costs, γ is the Phillips' Curve slope and ξ_p is the Calvo's price stickyness parameter
- The two models are observationally equivalent in terms of prices and quantities, the difference between the two models lies in the γ, not on ξ_p

ACEL2011's Mechanism (Cond't)

- In the **Homogeneous** capital model γ (eq. 5) depends on agent's discount rate β and on the fraction $1 - \xi_p$ of firms that can optimise within the quarter
- In the **Firm Specific** γ depends on a broader sets of structural parameters, is a non linear function with a non trivial representation.
- The more costly is for a firm to vary capital utilization, the steeper is its marginal cost curve and hence the smaller γ is
- The more elastic is the individual firm's demand curve, the smaller is γ

ACEL2011 Results

The estimated γ is 0.014, this implies that in:

- Homogeneous Capital model, firms change prices every 5.6 quarters, with an implied $\xi_p = 0.82$
- Firms Specific model, firms change prices every 1.5 quarters, with an implied $\xi_p = 0.33$

The reason why the models have such different implications for firms' pricing behaviour is that, firm's demand curve are highly elastic and their marginal curves are very steep
Only the Firm-specific capital model reconcile the micro-macro pricing conflict

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- **Are these results robust to an enriched lag structure of the VAR?**

VAR IRFs; Monetary Policy Shock

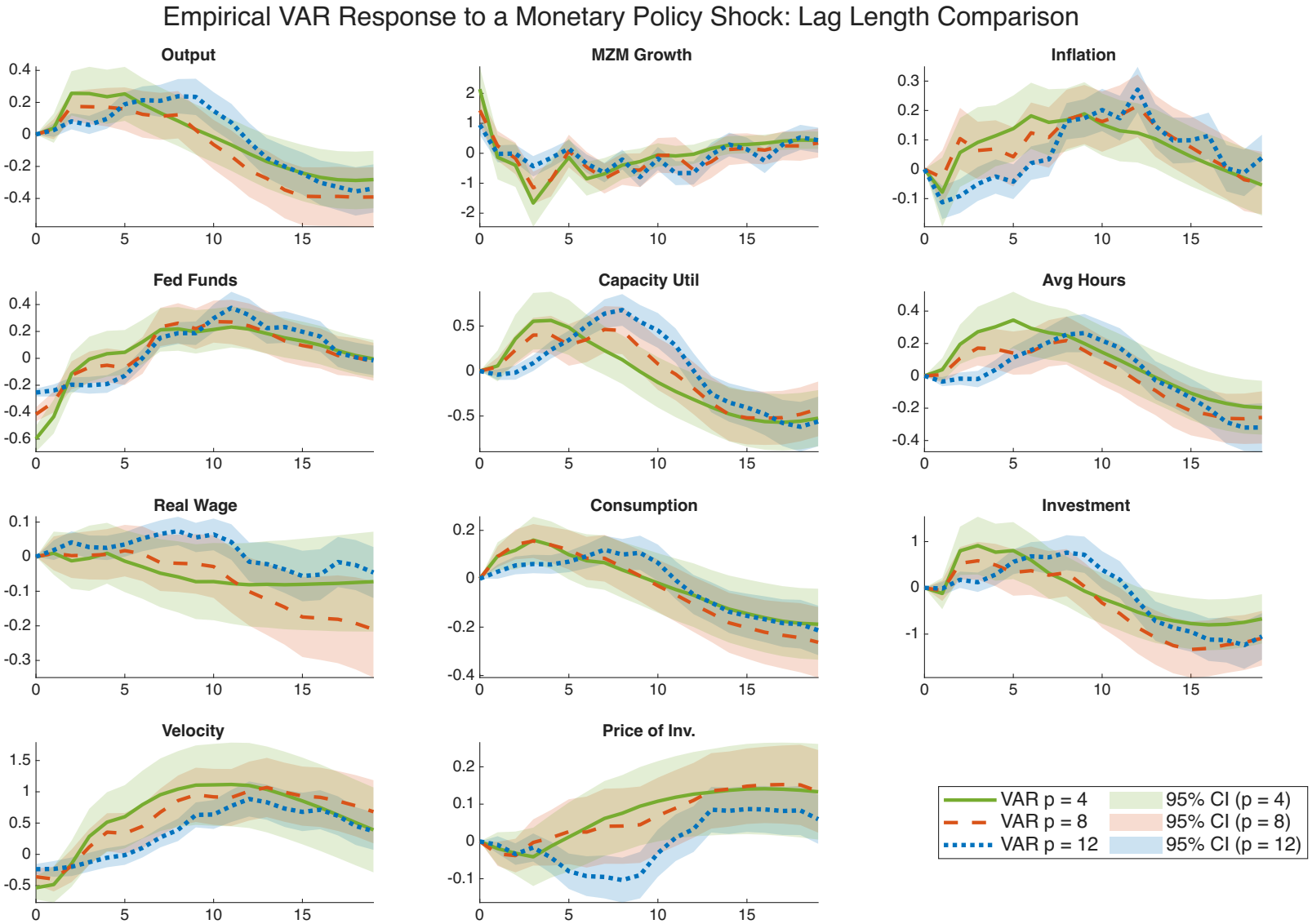


Figure 2: IRFs VAR Comparison: Monetary Policy Shock

DSGE IRFs; Monetary Policy Shock

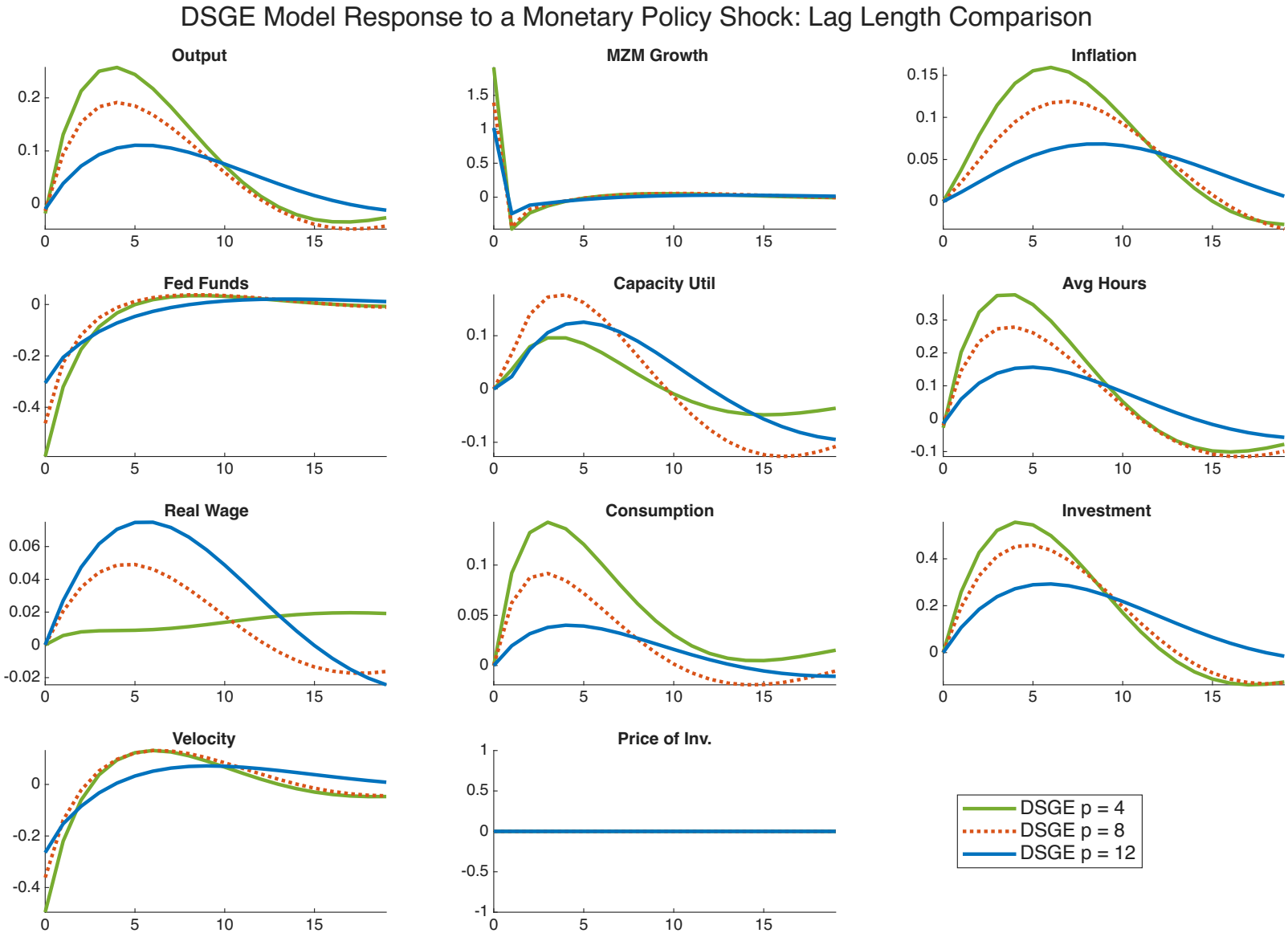


Figure 3: IRFs DSGE Comparison: Monetary Policy Shock

DSGE Structural Parameters

Parameter	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Structural Estimated Frictions			
Price markup (λ_f)	1.0100	1.0100	1.0100
	(-)	(-)	(-)
Wage stickiness (ξ_w)	0.7221***	0.5617***	0.4838***
	(0.1506)	(0.1142)	(0.0757)
Phillips curve slope (γ)	0.0402	0.0402	0.0129***
	(0.0246)	(0.0598)	(0.0042)
Capacity util. cost (σ_a)	1.9938	0.1175	0.0042
	(2.6066)	(0.6299)	(0.0230)
Habit formation (b)	0.7066***	0.6935***	0.8596***
	(0.0463)	(0.0570)	(0.0203)
Inv. adj. cost (S'')	3.2807*	3.0372*	6.8431***
	(1.8874)	(1.5506)	(2.3306)
Money demand elast. (ϵ)	0.8077***	0.7531***	0.8346***
	(0.2111)	(0.1473)	(0.1944)
Implied Price Stickiness			
Homogeneous capital ($\xi_{p,hom}$)	0.8213	0.8213	0.8958
	(-)	(-)	(-)
Firm-specific capital ($\xi_{p,fs}$)	0.3372	0.6448	0.8884
	(-)	(-)	(-)

Back

DSGE Structural Parameters All Lags

Parameter	**VAR** $p = 4$	**VAR** $p = 6$	**VAR** $p = 8$	**VAR** $p =$
	Structural Estimated Frictions			
Price markup (λ_f)	1.0100	1.0100	1.0100	1.0100
	(-)	(-)	(-)	(-)
Wage stickiness (ξ_w)	0.7221***	0.9269***	0.5617***	0.6918**
	(0.1506)	(0.0705)	(0.1142)	(0.1276
Phillips curve slope (γ)	0.0402	0.1519	0.0402	0.2827
	(0.0246)	(0.2283)	(0.0598)	(0.3277
Capacity util. cost (σ_a)	1.9938	10.5446	0.1175	0.0853
	(2.6066)	(27.8459)	(0.6299)	(0.2326
Habit formation (b)	0.7066***	0.7183***	0.6935***	0.7802**
	(0.0463)	(0.0573)	(0.0570)	(0.0407
Inv. adj. cost (S'')	3.2807*	2413.4944	3.0372*	10.0700
	(1.8874)	(7400.6891)	(1.5506)	(7.2363
Money demand elast. (ϵ)	0.8077***	0.8089***	0.7531***	0.9409**
	(0.2111)	(0.1996)	(0.1473)	(0.2900
	Implied Price Stickiness			
Homogeneous capital ($\xi_{p,hom}$)	0.8213	0.6809	0.8213	0.5928
	(-)	(-)	(-)	(-)
Firm-specific capital ($\xi_{p,fs}$)	0.3372	0.1055	0.6448	0.3655
	(-)	(-)	(-)	(-)

Implied Average Time between Re-Optimization

Table 1: Implied Average Time between Re-Optimization

Model Specification	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Homogeneous Capital Model			
Calvo parameter ($\xi_{p,hom}$)	0.82	0.82	0.90
Implied duration (quarters)	**5.6**	**5.6**	**9.6**
Firm-Specific Capital Model			
Calvo parameter ($\xi_{p,fs}$)	0.34	0.64	0.89
Implied duration (quarters)	**1.5**	**2.8**	**9.0**

- **Baseline VAR** ($p = 4$): Homogeneous capital implies price changes every **5.6 quarters**.
- **Firm-specific capital**: Drastically reduces this to **1.5 quarters**, matching microeconomic evidence.

Note: Implied duration is calculated as $D = 1/(1 - \xi_p)$.

Implied Average Time between Re-Optimization

Table 2: Implied Average Time between Re-Optimization

Model Spec.	**VAR** $p = 4$	**VAR** $p = 6$	**VAR** $p = 8$	**VAR** $p =$
Homogeneous Capital Model				
Calvo ($\xi_{p,hom}$)	0.82	0.68	0.82	0.59
Duration (quarters)	**5.6**	**3.1**	**5.6**	**2.5**
Firm-Specific Capital Model				
Calvo ($\xi_{p,fs}$)	0.34	0.11	0.64	0.37
Duration (quarters)	**1.5**	**1.1**	**2.8**	**1.6**

- **Baseline VAR** ($p = 4$): Homogeneous capital implies price changes every **5.6 quarters**.
- **Firm-specific capital**: Drastically reduces this to **1.5 quarters**, matching microeconomic evidence.

Note: Implied duration is calculated as $D = 1/(1 - \xi_p)$.

Next Step

In the ACEL2011 exercise

- MonteCalvo of the Phillipcs curve slope components to simulate the Calvo Price Parametes' Standard Error
- Check how good distance between IRFs is minimized
- Check the estimated parameters likelihood function
- Leverage on Canova's tip; long-lag VARs mess up HD
- Long-Lag VARs are a good aproximation of Trend-Cycle VARs, are we specifying better the trend?
- Make sure the VARs starts from the same point in time, we need to drop the first 12 data points fro each VAR for comparison pourposes

Future Research

- Same exercise using Jarociński and Karadi2020)
- Analyse the Long-Lag VAR results in both the Frequentist and Bayesian approach

Thanks a lot for your attention!

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VAR IRFs; Monetary Policy Shock

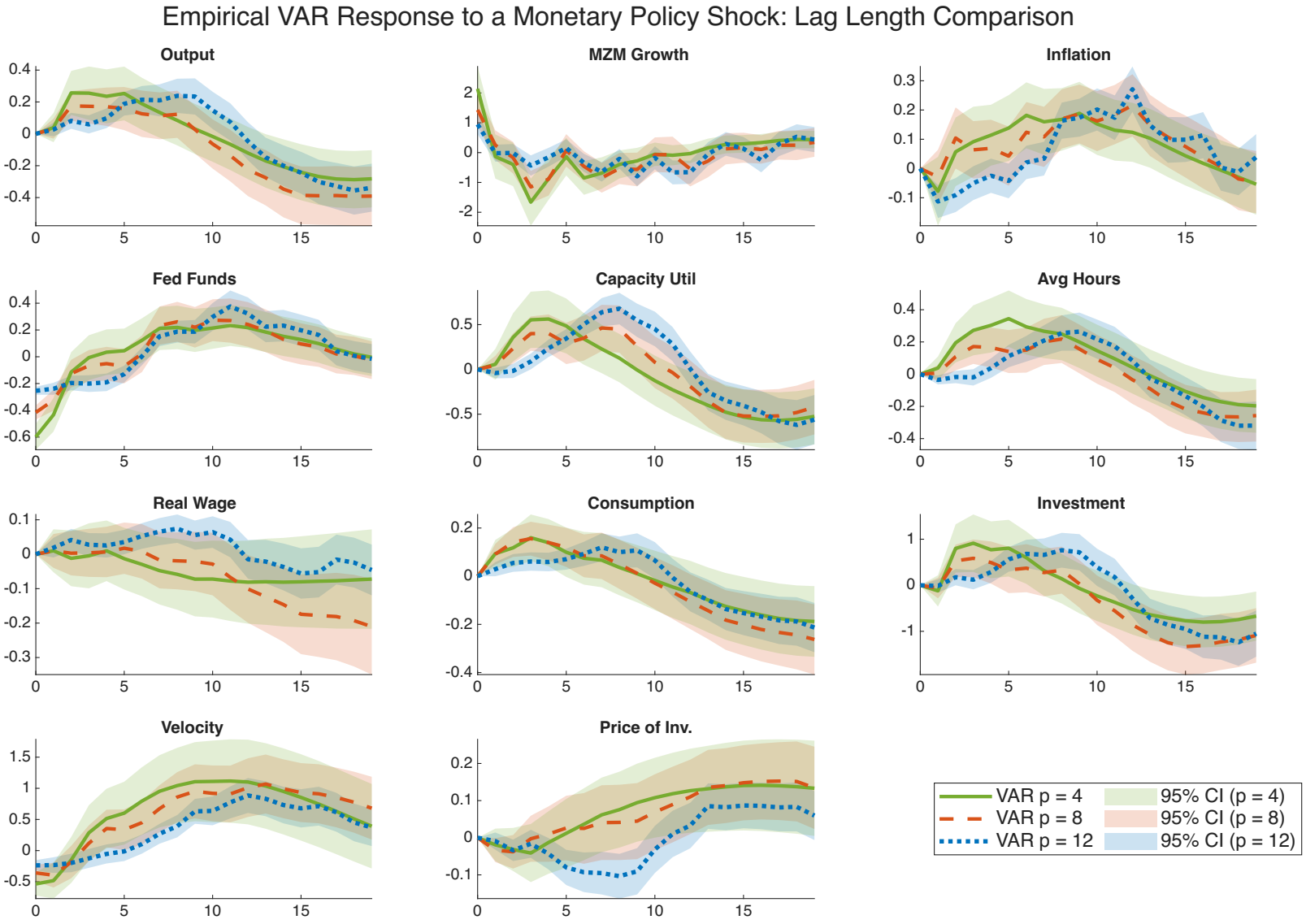


Figure 4: IRFs VARs all CI Comparison: Monetary Policy Shock

Monetary Policy Shock; DSGE4 vs VAR4

Figure 1: Response to a monetary policy shock (o - Model, - VAR, grey area - 95 % Confidence Interval)

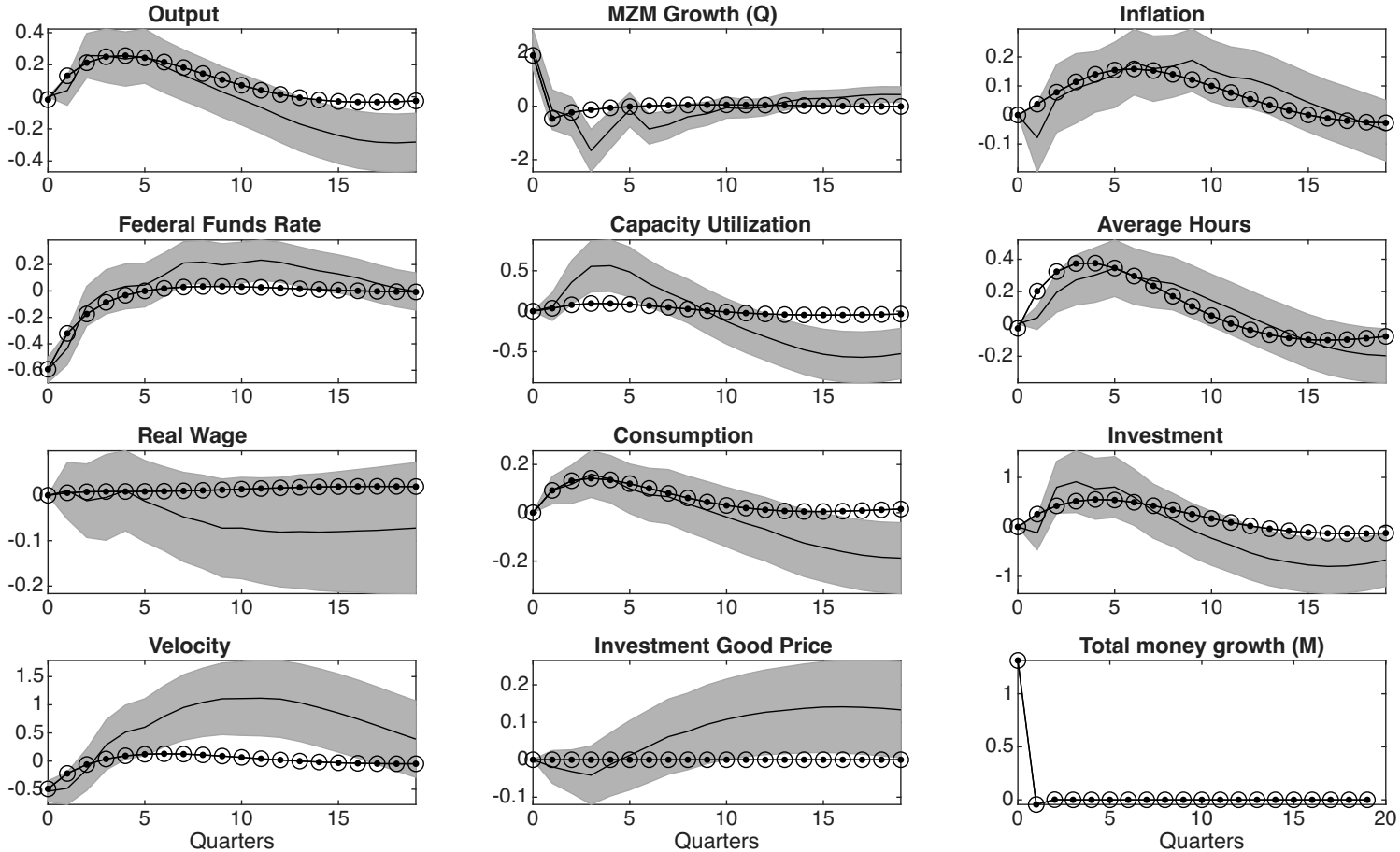


Figure 5: MP: DSGE4 Vs VAR4

Monetary Policy Shock; DSGE4 vs VAR8

Figure 1: Response to a monetary policy shock (o - Model, - VAR, grey area - 95 % Confidence Interval)

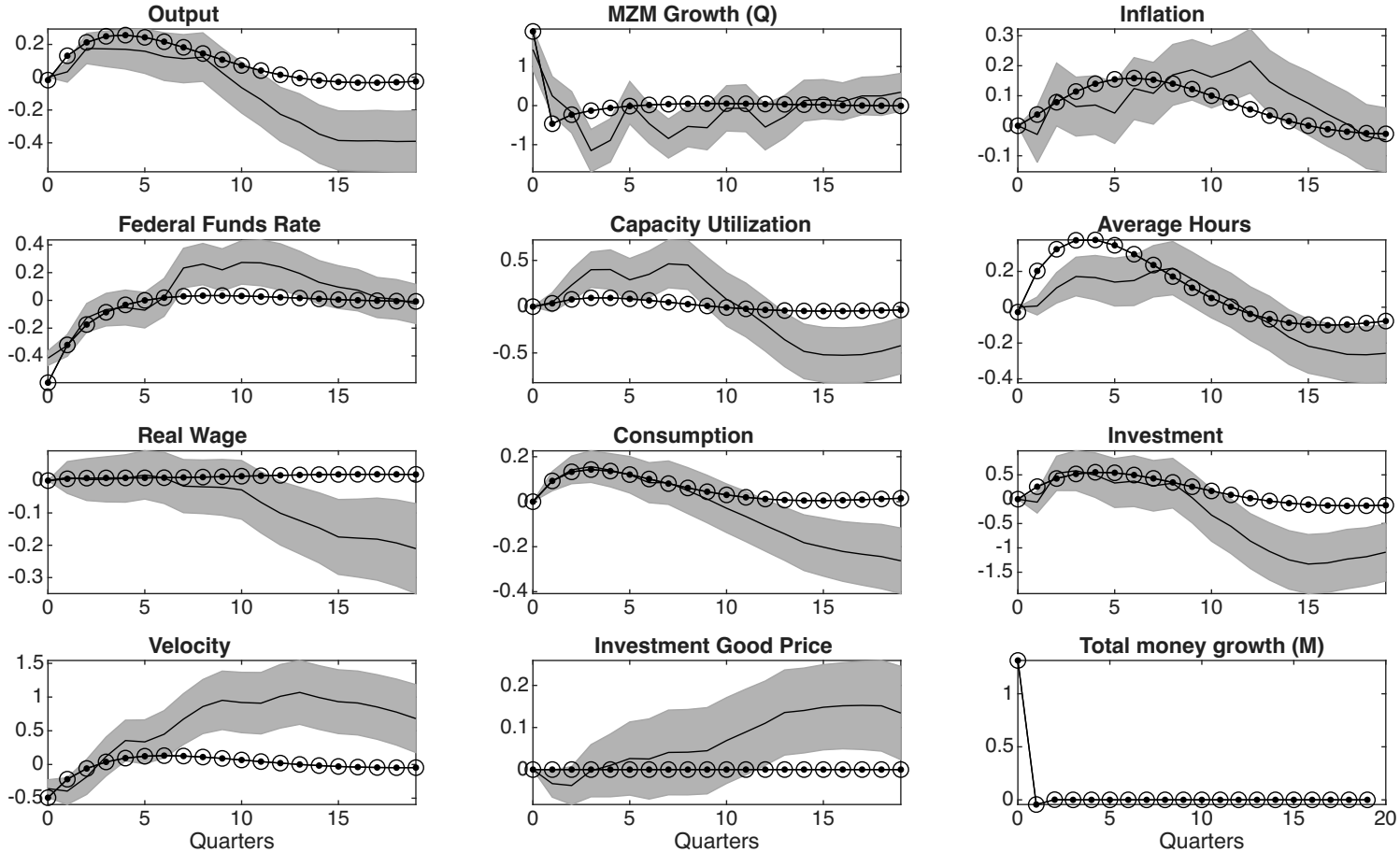


Figure 6: MP: DSGE4 Vs VAR8

Monetary Policy Shock; DSGE4 vs VAR12

Figure 1: Response to a monetary policy shock (o - Model, - VAR, grey area - 95 % Confidence Interval)

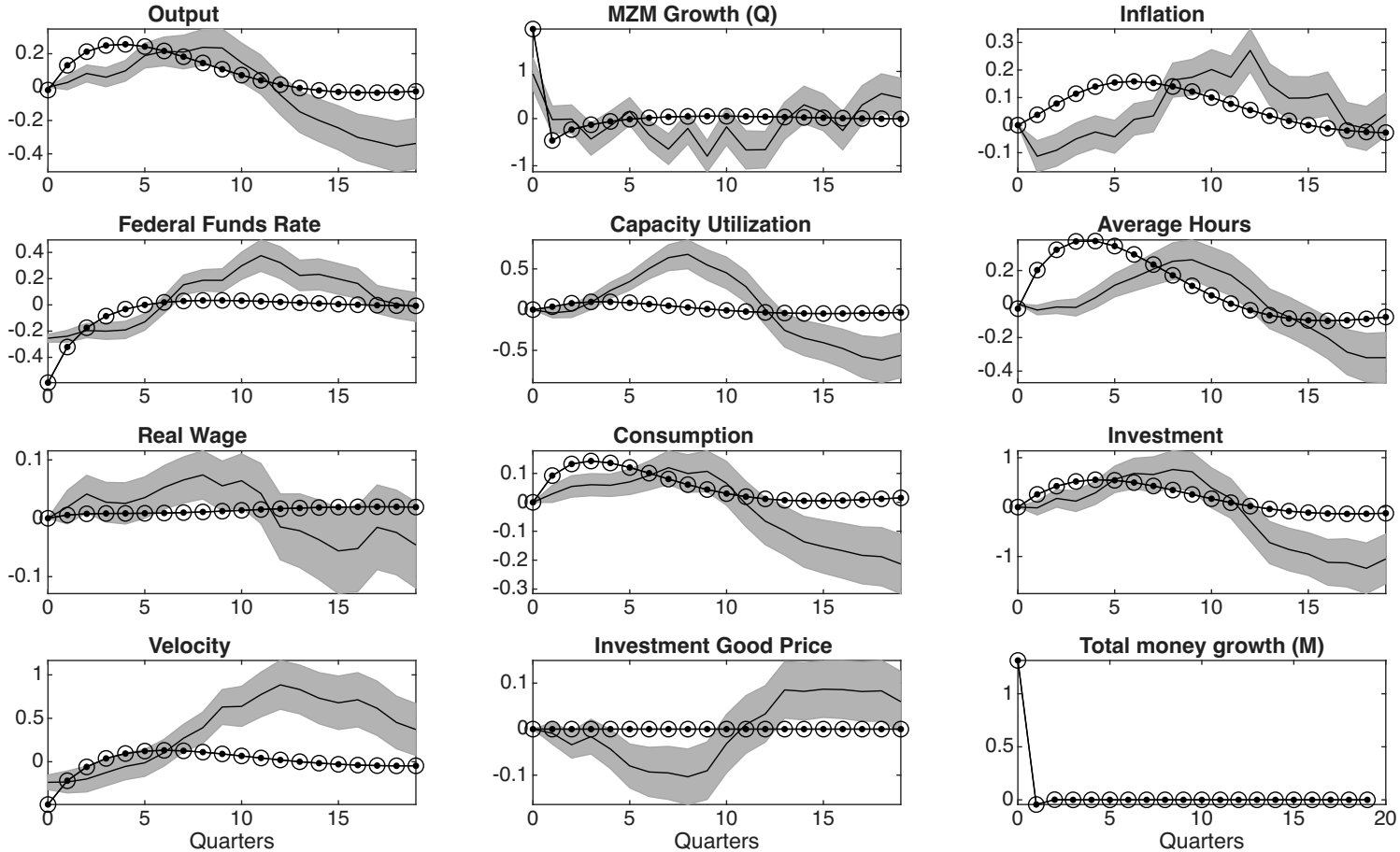


Figure 7: MP: DSGE4 Vs VAR12

VAR IRFs; Neutral Technology Shock

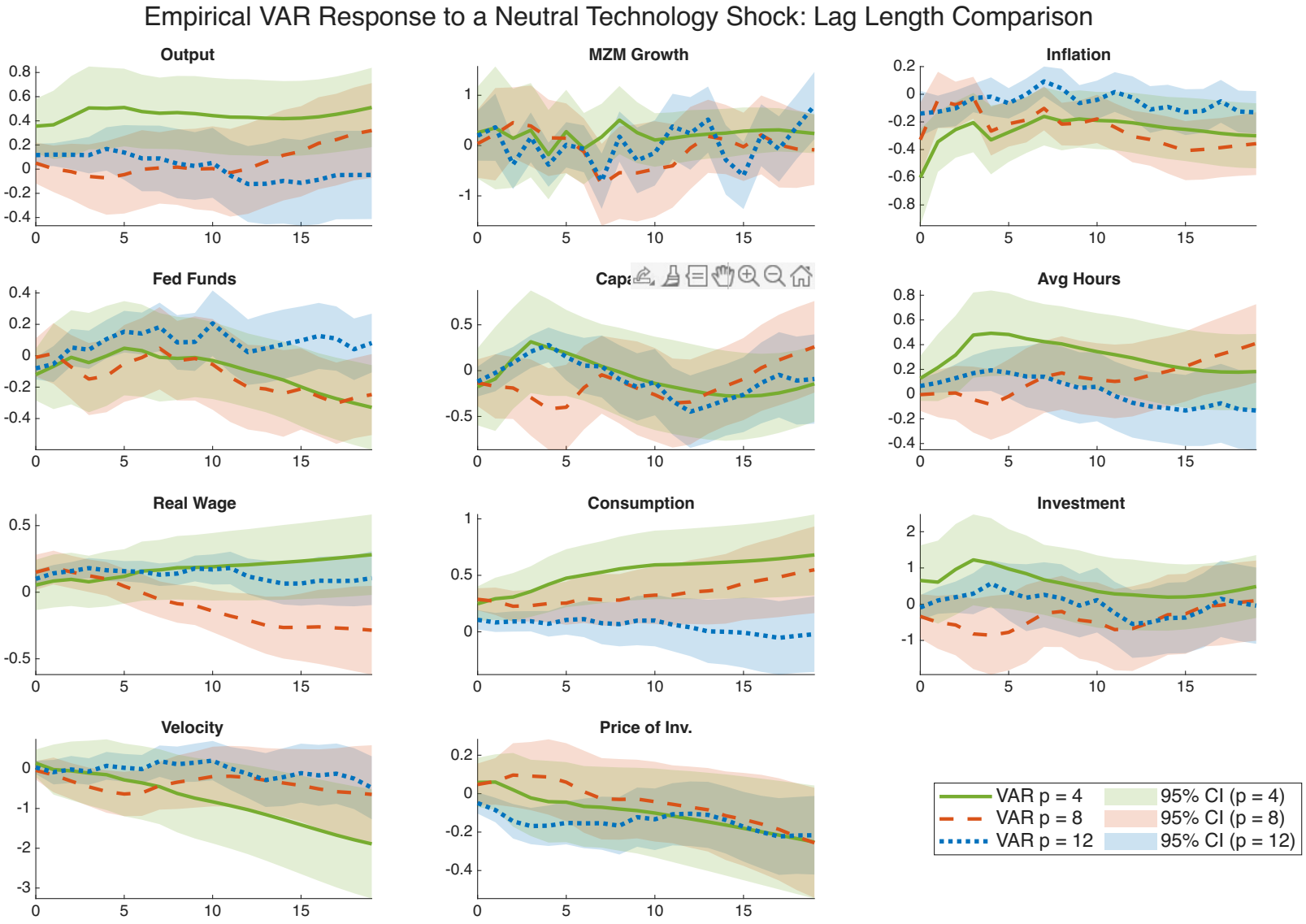


Figure 8: IRFs VARs all CI Comparison: Neutral Technology Shock

DSGE IRFs; Neutral Technology Shock

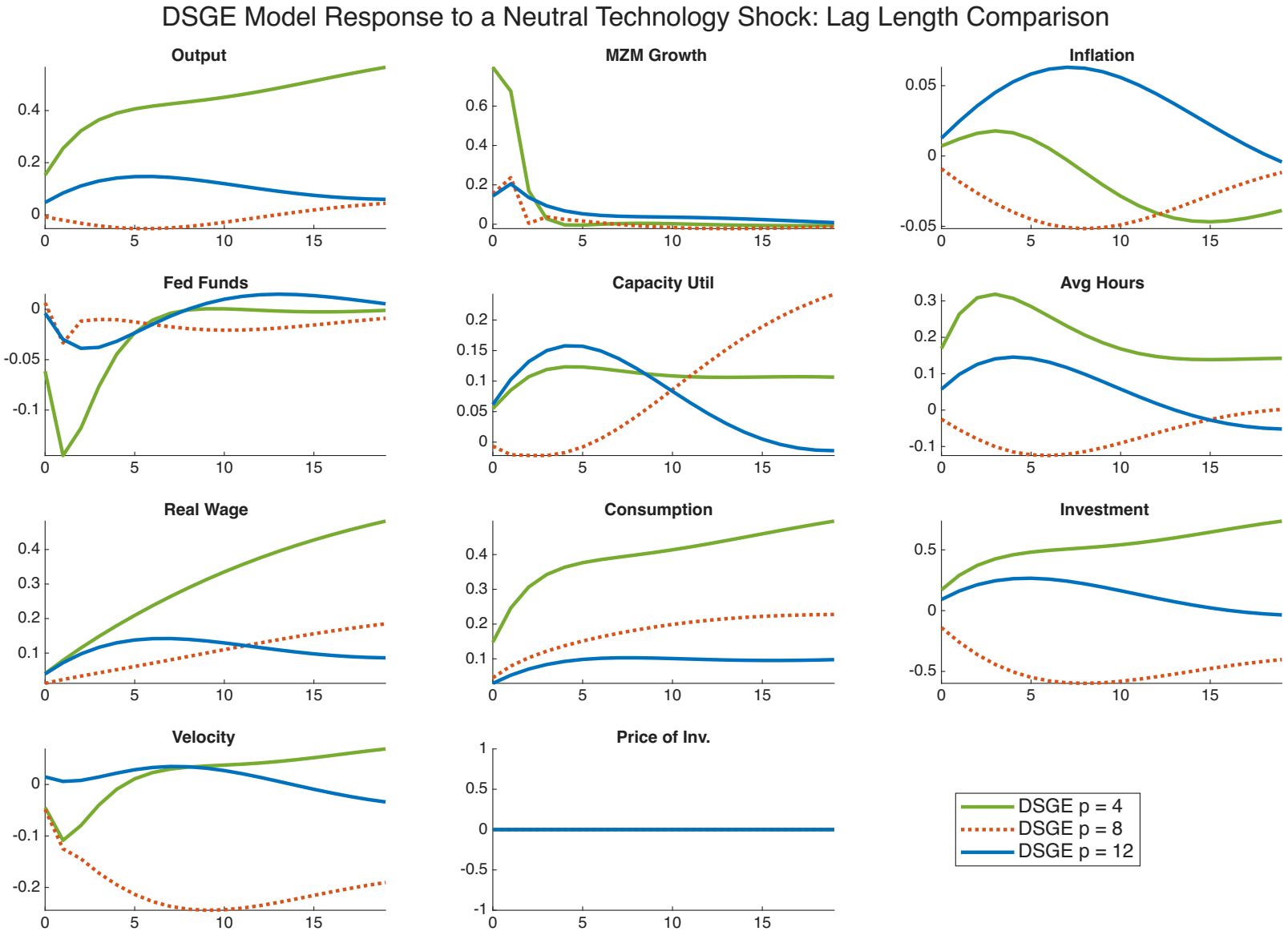


Figure 9: IRFs DSGE Matched/IRF Comparison: Neutral Technology Shock

Neutral Technology Shock; DSGE4 vs VAR12

Figure 2: Response to a neutral technology shock (o - Model, - VAR, grey area - 95 % Confidence Interval)

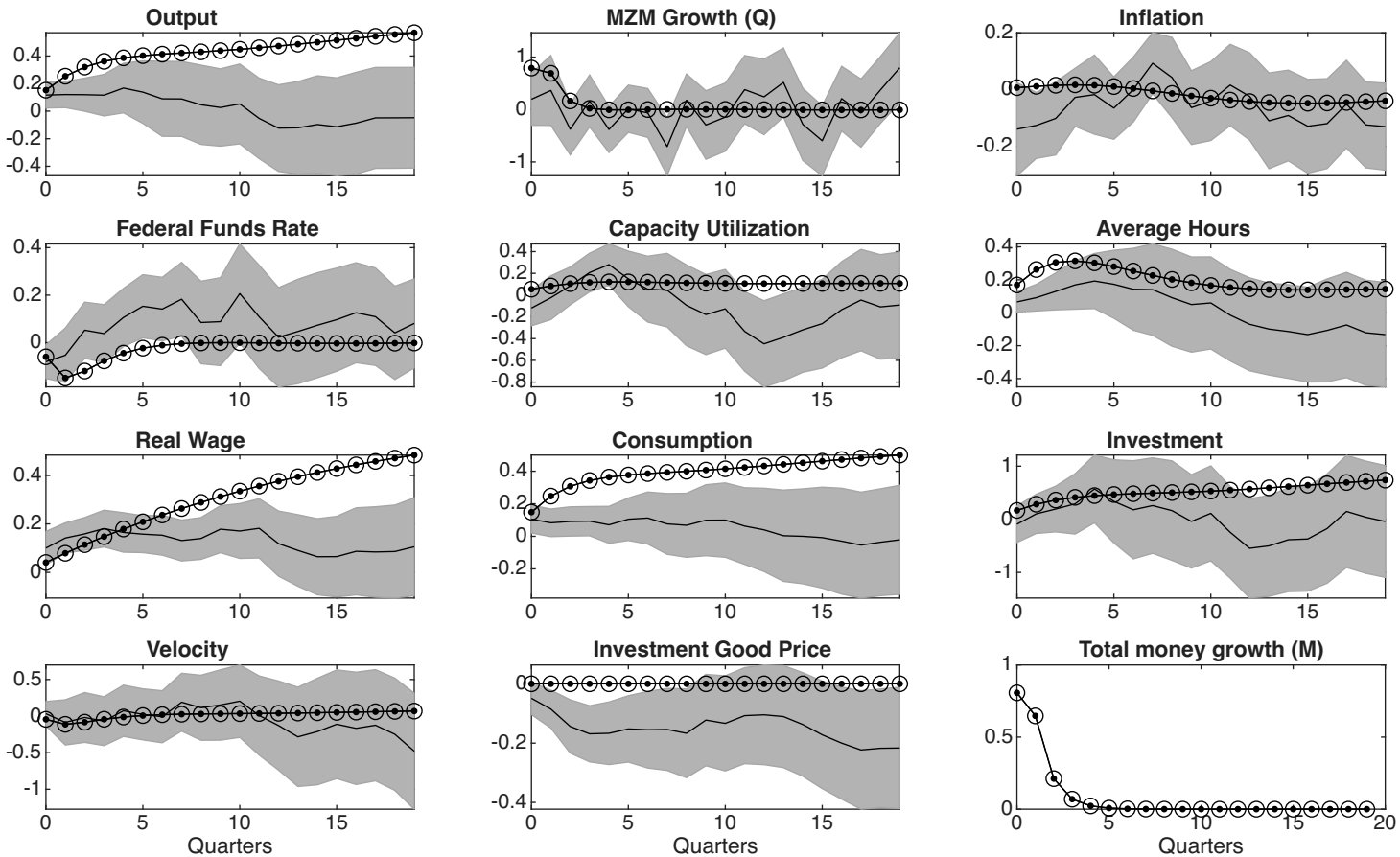


Figure 10: NT: DSGE4 Vs VAR12

Neutral Technology Shock; DSGE12 vs VAR12

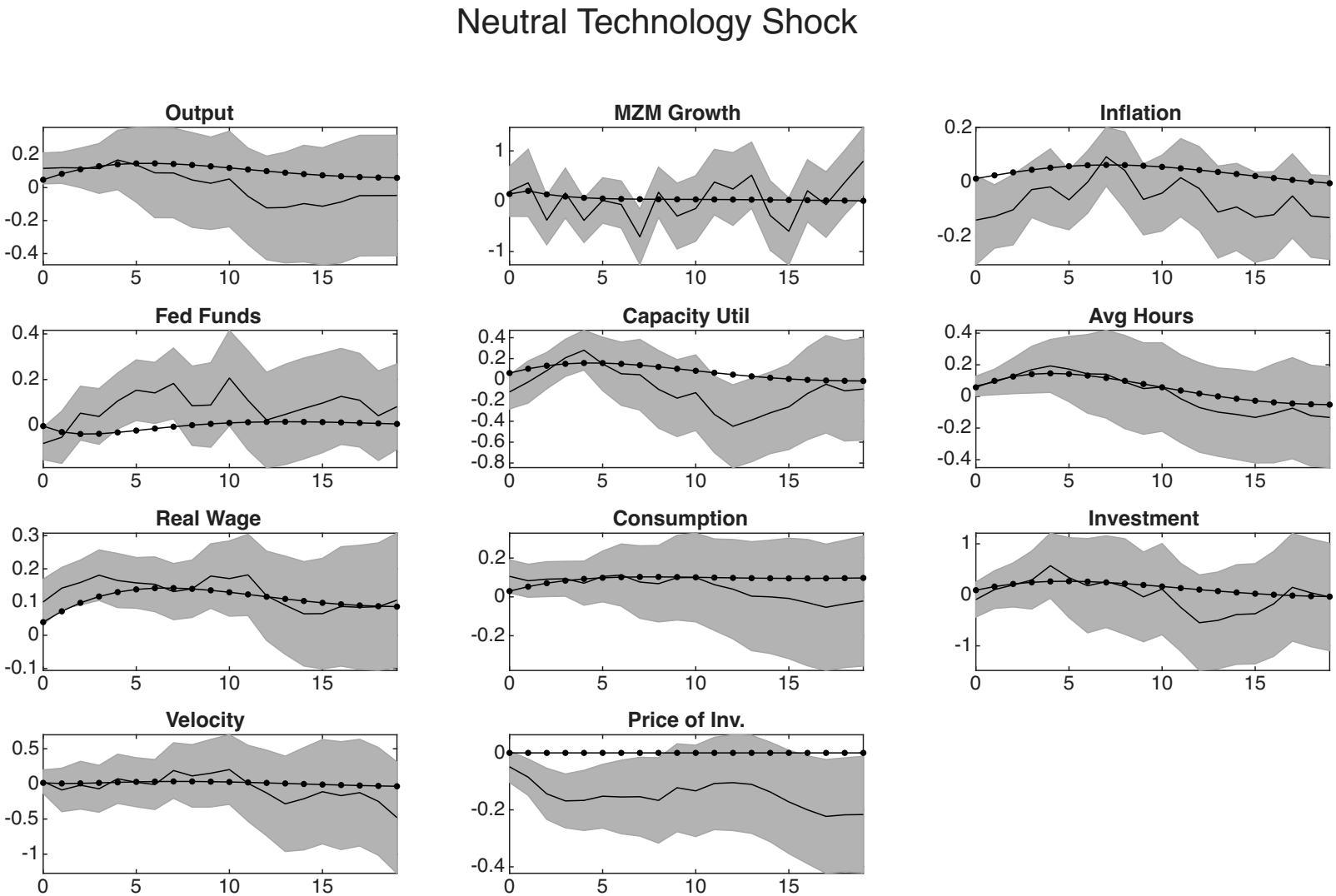


Figure 11: NT: DSGE12 Matched with VAR12

VAR IRFs; Embodied Technology Shock

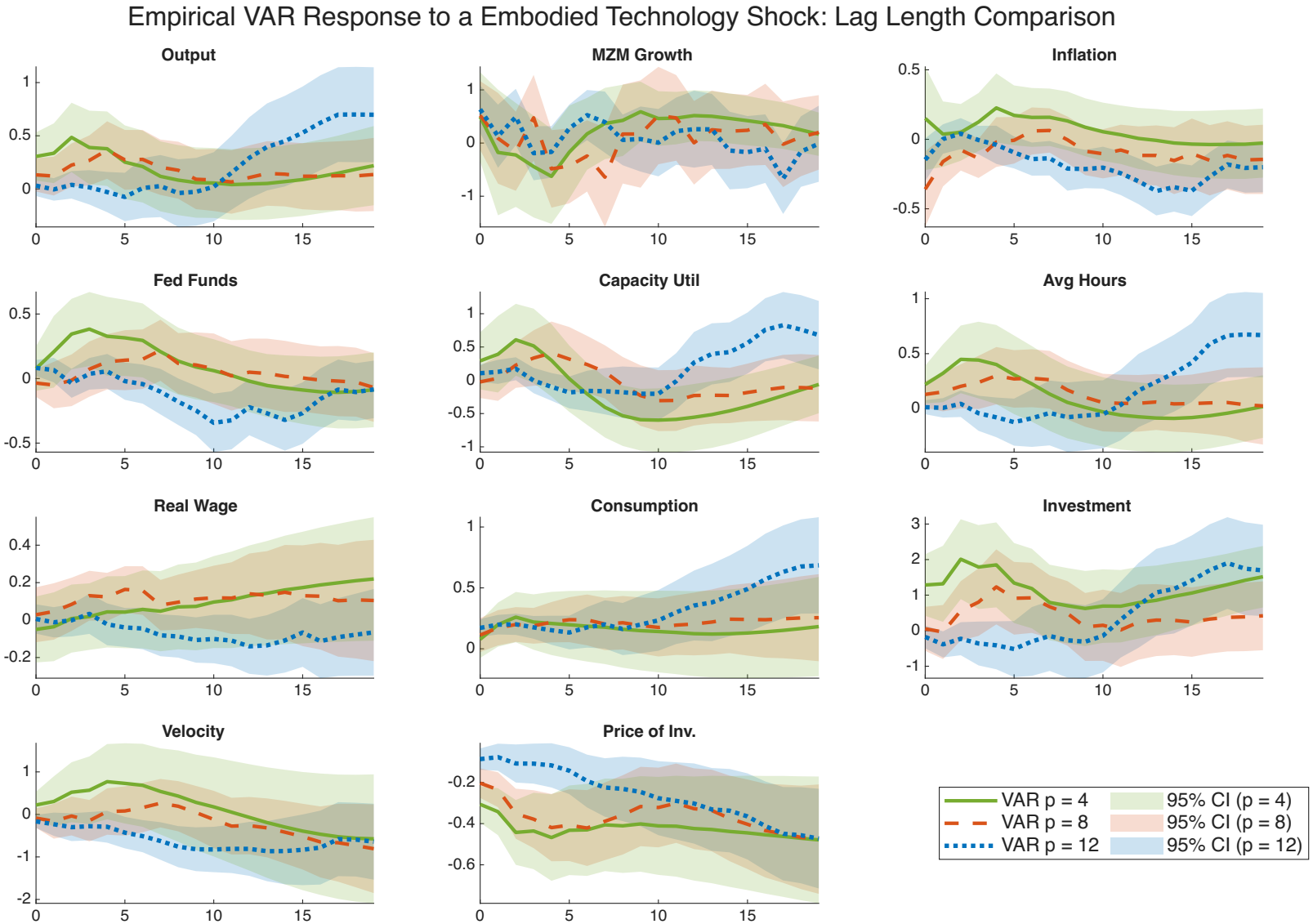


Figure 12: IRFs VARs all CI Comparison: Embodied Technology Shock

Embodied Technology Shock; DSGE4 vs VAR12

Figure 3: Response to an embodied technology shock (o - Model, - VAR, grey area - 95 % Confidence Interval)

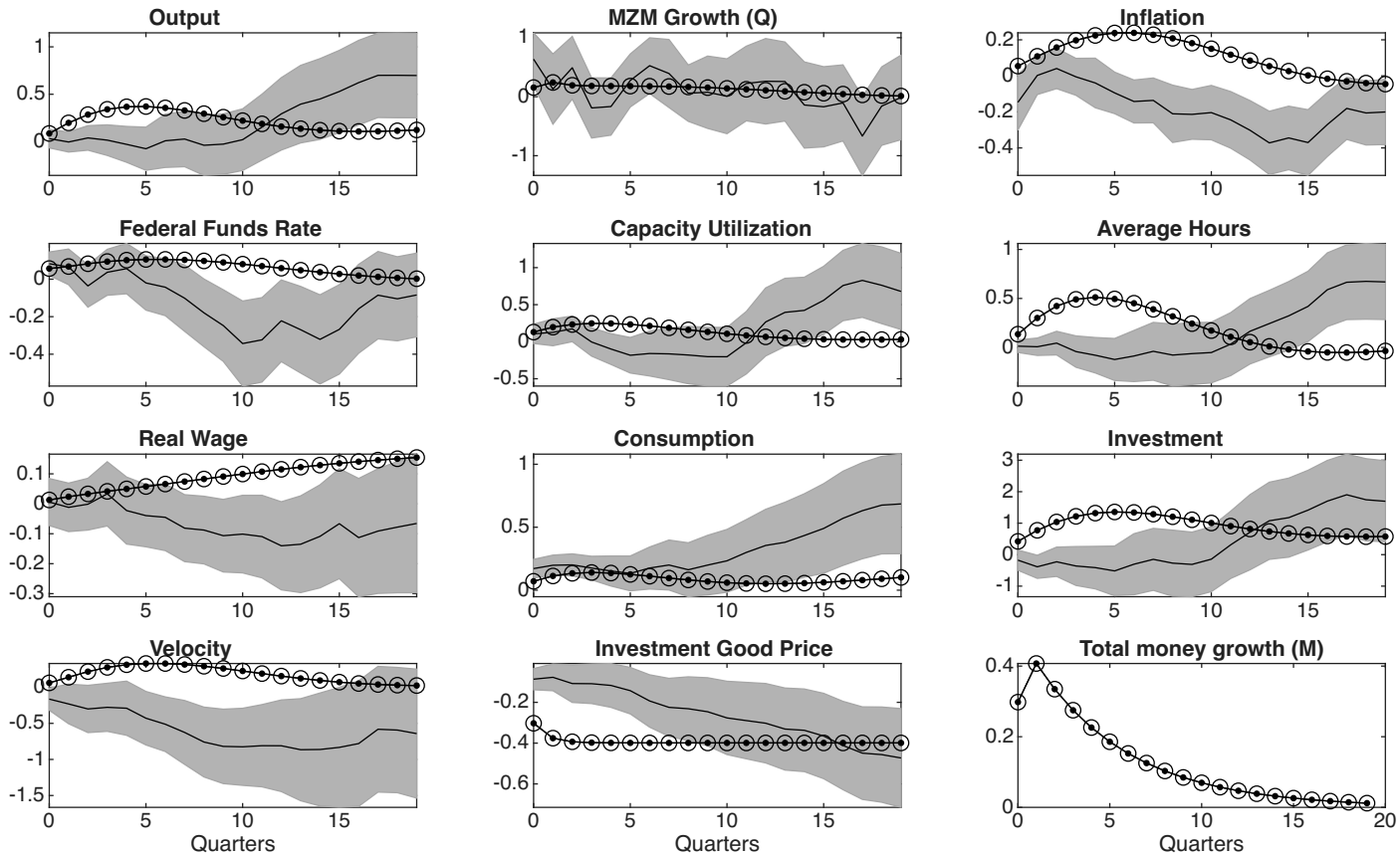


Figure 13: ET: DSGE4 Vs VAR12